COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

192511347

I Shanmuganathan SRS *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

SSTG

Signature of the student:

20 + 20 + 25 + 17 + 10

= 92

100

Uniform cost Search (UCS):

Uniform Cost Search expand the node with the lowest Cumulative path cost from the start node (A)

Graph Brakedown:-

A to B : 2

A to E : 4

B to D : 5

B to E : 1

C to E : 2

E to D : 3

Priority Queue Trace Table:

Step	Priority Queue	Expanded	Cost
1	[(A,0)]	A	0
2	[(B,2),(E,4)]	B	2
3	[(E,3),(D,7)]	E	3
4	[(D,6)]	D	6
5	[ ]	–	–

Least cost path: A → B – E – D

Total cost : 6

Problem formulation

* Initial state : At the college gate
* Goal state : At the CSE Laboratory
* State space : Set of all location [Gate, Block A, Library, Block B, Canteen, CSE Lab]
* Action / operations : Move to an adjacent connected location (e.g. Move (Gate, Block A), Move (Block A, Library), Move (Gate, Canteen), etc).
* Path cost : Sum of individual distance or travel time between location along the chosen route.
* One possible solution path : Gate -> Block A -> Library ->

Intelligent Agent and Rationality.

A rational agent select action that maximize performance measure based on its percept sequence. r a vacomn Cleaner the goal is to maximize anliness while minimizing unnecessary movement

Rational Agent table:-

Location	Room status	Rational Action
A	Dirty	Suck
A	Clean	move right
B	Dirty	Suck
B	Clean	move Left

Justification of the Agent's

* When A room is Dirty:- The most rational action is Suck because it directly cleans the environment maximizing the performance metric

* When a room is clean:-

Sucking is redundant. The rational choice is to move ~~to~~ the other room to check for dirt ensuring the entire environment because or stays clean.